

Non-invasive bat species identification from mixed-species samples using a microarray

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Abstract Identifying species from non-invasively collected, mixed species samples for biodiversity monitoring can be difficult and expensive using typical molecular methods because samples are often degraded. This is the case when identifying the species of bats present in roosts where guano may be the only means of assessment without disturbing the bats themselves. To aid in such studies, we developed species-specific DNA probes and a microarray capable of identifying most bat species in the United States and Canada using our existing database of 16S mitochondrial DNA sequences. The microarray was sensitive enough to detect DNA diluted 1:500 for several species combinations and was able to detect the presence of more species in mixed guano samples collected from roosts than did direct sequencing from individual fecal pellets. We suggest that these DNA probes and the microarray could be a valuable tool with which to non-invasively monitor bat populations and roost use.

Keywords Microarray · Bat · Guano · Species identification · Non-invasive sampling · Mixed species samples · Degraded DNA

Introduction

A prerequisite for organismal and biodiversity studies is the ability to accurately identify species of interest and distinguish target species from all other possible alternatives. For most groups of animals and plants, species identification has traditionally relied on the use of morphological characters to distinguish members of different species. The use of genetic markers provides an alternate source of diagnostic characters to aid in species discrimination and is particularly useful in cases where morphology may be misleading or when species are elusive. Such cases include when there is cryptic morphological variation within a taxon or when using non-invasively collected samples such as hair or feces that cannot be directly attributed to a particular individual and may contain DNA from multiple individuals (Waits and Paetkau 2005).

Molecular species identification of vertebrates has typically focused on the direct sequencing of mitochondrial DNA (mtDNA) markers, due to their high mutation rate, lack of recombination, and high copy number within cells (Waits and Paetkau 2005; Sarri et al. 2014; Grattarola et al. 2015). Direct sequencing of partial or entire mtDNA gene fragments (typically >600 bp) can provide a sufficient number of informative sites to overcome intraspecific variation even when delineating closely-related species and provide data that can be used in downstream population genetic or phylogenetic analyses (Hajibabaei et al. 2006; Kekkonen and Hebert 2014). However, both Sanger and next-generation sequencing may be expensive when large numbers of

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individual samples are targeted and may not be necessary when the goals of the study require only knowledge of species presence or absence as is the case in many biodiversity monitoring programs.

A feasible alternative to direct sequencing for biotic inventories is the use of microarrays (e.g., Pfunder et al. 2004; Hajibabaei et al. 2007; Kochzius et al. 2008; Teletchea et al. 2008; Lin et al. 2014; Noyer et al. 2015). A microarray is a series of nucleic acid probes immobilized on a solid substrate which are complementary to gene fragments of a particular target. Fluorescently-labeled DNA or RNA from a test sample is applied to the array and binds to matching probes, providing a signal of presence/absence and relative concentrations of particular targets (genes or species) within a sample. Microarrays have traditionally been used in gene expression studies but have also been applied to species identification in a wide range of contexts, including pathogen surveillance (Cleven et al. 2006; Huang et al. 2006, 2013; Paulin et al. 2014), microbial ecology (Volokhov et al. 2003; Peplies et al. 2006; Bonhomme et al. 2007; Kellogg et al. 2012), and dietary supplement product verification (Carles et al. 2005; Patro et al. 2015). The major advantages of microarrays for molecular species identification are their low production costs and the fact that they can be used to simultaneously identify hundreds or thousands of targets within a sample. Thus, researchers interested in detecting the presence of a particular taxonomic group can prepare a microarray with multiple and redundant species-specific probes to identify all contributors to a mixed-species sample in a single reaction. In addition, the sensitivity of microarrays allows for the detection of small quantities of DNA and subtle differences among samples that are much harder to detect with other molecular methods (Bryant et al. 2004; Forêt et al. 2007), making them ideally suited for use with non-invasively collected samples with low DNA content. In turn, this could mean significant savings of time and resources when simultaneously screening a high volume of samples for accurate identification of multiple species.

Microarray probes may be designed for any marker or set of markers and the choice of which gene(s) to use will determine the functionality of the array. Pfunder et al. (2004) provided a ‘proof-of-concept’ for the use of microarrays based on partial COI sequences for mammalian species identification and advocated the development of a ‘biodiversity chip’ for diversity monitoring. Hajibabaei et al. (2007) compared direct sequencing of large fragments of mitochondrial genes (COI and Cytochrome *b*) and a microarray based on probes from these same genes for mammals. They concluded that both direct sequencing and microarrays provide accurate species identification, but suggested that microarrays have the drawback of requiring advance knowledge of the target sequences so that probes matching all variants within

a species can be included on the array. Microarrays are increasingly being applied to biodiversity studies of animals or plants. Recently, Lung et al. (2013) also used the COI gene to develop a microarray capable of identifying Canadian bat species. However, the markers used in these studies (≥ 650 bp) are much longer than the short DNA fragments typically retrieved from non-invasive samples such as feces, hair, or museum specimens (< 200 – 300 bp; Taberlet et al. 1999; Waits and Paetkau 2005), limiting their broad applicability. Although smaller regions of these genes theoretically may provide sufficient variation for species delimitation (Hajibabaei et al. 2006), variation in coding DNA tends to be relatively evenly distributed across the gene and saturated in third codon positions (Pesole et al. 1999). This renders the development of species-specific probes and conserved primers for amplifying the short regions necessary for degraded samples highly problematic in practice. By contrast, the ideal markers used to develop microarray probes for species identification should be short to accommodate samples of varying quality, flanked by highly conserved sequence to facilitate the development of universal primers, and contain fixed variation for distinguishing species. Such characteristics are typical of ribosomal genes, which often have conserved sequences flanking highly variable regions of secondary structure containing numerous point mutations and indels (Hillis and Dixon 1991) across which probes can be designed with high specificity (Salathia et al. 2007; Kochzius et al. 2010).

Here we report on the development of species-specific DNA probes and a microarray capable of identifying nearly all of the 47 bat species found in the USA and Canada using non-invasively collected fecal samples. Bats are an ideal target group for microarray-based molecular species identification because they are elusive and difficult to survey effectively using capture methods and field surveys, and cryptic morphological variation among closely-related species is common (Mayer and Helversen 2001; Ruedi and Mayer 2001; Stadelmann et al. 2007). Furthermore, non-invasive sampling of bat feces at roost sites is an efficient, cost-effective means of detecting bat presence, but requires an alternate means of species identification. Our microarray is based on two short (190 bp) sections of the mitochondrial 16S ribosomal subunit gene containing clusters of interspecific variability (Zinck et al. 2004). We outline the steps in probe design and production of the microarray and demonstrate 100 % classification success in test hybridizations of both single and mixed species samples. Molecular analyses that utilize non-invasive sampling methods allow us to detect species where previously it was not possible to do so, and make it feasible to assess species composition over time in a way that provides a different ecological perspective than is provided using traditional capture and release survey methods.

Methods

Probe design

We used our database of sequences from two fragments of the 16S ribosomal subunit gene (Zinck et al. 2004) from the majority of bat species in the United States and Canada (792 individuals, 39 species) to design and evaluate probes for the microarray. We treated three subspecies of *Myotis lucifugus* (subspecies *alascensis*, *carissima*, and *lucifugus*) separately with respect to probe design as they are easily distinguishable based on mtDNA sequences (Vonhof et al. 2015) and identification of subspecific identity was important to monitoring goals in parts of western North America, resulting in 41 named taxa being included. That said, although the two 16S fragments are sufficient to differentiate among most of the bat species in the US and Canada, there are unresolved species complexes within which we are unable to distinguish between putative species (Table 1 and Zinck et al. 2004) and have proven difficult for others to identify using a variety of molecular markers (e.g., Rodriguez and Ammerman 2004; Carstens and Dewey 2010). Combining these species into operational units resulted in a total of 30 taxonomic units included on the array (Table 1). The sequences in our database were obtained following protocols outlined in Zinck et al. (2004) with additional sequences acquired from GenBank (Table 1). Probe design for six species relied on GenBank sequences alone (Table 1). We were unable to design probes for *Eumops glaucinus*, *E. perotis*, *E. underwoodi*, *Leptonycteris nivalis*, *Macrotus californicus*, and *Myotis occultus* due to lack of genetic material or sequences on GenBank.

To ensure that probes were responsive to the range of genetic variability within a species, DNA from an individual bat representing each known haplotype per species was included in an alignment from which oligonucleotide probes of between 16 and 74 base pairs were manually designed. Haplotypes were identified using intraspecific p-distances (Zinck et al. 2004). Although we feel that we have a reasonable sample of the genetic diversity of bats across the US and Canada, we recognize that there may be geographic gaps in our samples. Because our target fragment was <200 base pairs, probes were not eliminated on the basis of GC content or melting temperature. Probes were initially ordered with a 5'-C6-aminoterminator and were normalized to 100 µM in TE (Tris–EDTA) pH 8.0.

Microarray design

Corning® Epoxide™ Coated Slides were used for all trials detailed below. A 384 well plate was prepared with 1.5 µl of 100 µM probe, 6 µl ddH₂O, and 7.5 µl 6X Print Buffer [300 mM Sodium Phosphate (pH 8.5)]. Probes were

spotted three times onto each slide with positive controls in each block to assist in positioning of the array layout for analysis. Slides were printed using a BioRobotics Microgrid (Genomic Solutions), held for 30 min, then transferred into a hydration chamber (saturated NH₄Cl + saturated KNO₃) for 12–17 h. Slides were then stored in a desiccator. For spotting quality control, microarrays were scanned with a Perkin Elmer ScanArray 4000 scanner.

DNA extraction and amplification

Guano samples were desiccated after collection through air- or oven-drying or exposure to a desiccant (Drierite or silica beads) prior to storage. DNA extraction and amplification for individual guano pellets followed the protocol in Zinck et al. (2004). Typically, DNA extraction and amplification from guano yielded 21 ng/µl of DNA. Mixed guano samples were homogenized prior to DNA extraction by manually crushing and mixing guano pellets using sterile instruments. DNA was then extracted from two to three 0.3 g samples for every 10 g of homogenized mixed-species sample using a modified Qiaquick PCR Purification (Qiagen) protocol. This kit has been shown to successfully extract amplifiable DNA from very degraded samples without the need for fume hoods (necessary for phenol/chloroform extractions) or specialized equipment (Yang et al. 1998). We added 0.3 g of homogenized guano, 1.35 ml ATL, and 50 µl proteinase K to a 1.5 ml microcentrifuge tube and incubated the tube at 55 °C overnight. Tubes were then centrifuged for 10 min. The supernatant was added to a new 1.5 ml tube and centrifuged for 5 min. The supernatant was divided between 3 new 1.5 ml tubes (approx. 200 µl each) and 1 ml of buffer PB was added to each tube. We added 750 µl of the supernatant/PB solution to a Qiaquick PCR Clean-up column (Qiagen) and centrifuged for 1 min. The flow-through was discarded and this step was repeated until all 3.6 ml of the supernatant/PB solution passed through the column. We then added 750 µl of buffer PE and centrifuged for 1 min, discarded the flow-through, and centrifuged for an additional minute. DNA was eluted in 50 µl of buffer EB.

PCR amplification followed the protocol in Zinck et al. (2004). Amplified DNA was cleaned using the Qiaquick PCR Clean-up protocol (Qiagen) and was then amplified a second time using labeled primers (Mysp1 and Mysp3 were labeled with a 5'Biotin modification, and primers Mysp2 and Mysp4 were labeled with a 5'Cy3 modification). To eliminate the possibility of preferential annealing of complimentary strands of the PCR amplicons (resulting in fewer amplicons hybridizing to the microarray probes) we removed one strand of the amplified PCR product using Dynabeads® M-270 Streptavidin magnetic beads. Post-PCR, the protocol for the Dynabeads® M-270 Streptavidin magnetic beads was followed to bind the strand with the

Table 1 Bat species investigated in this study and their abbreviations

Species	Abbreviation	Sequences from GenBank	Mysp1/2		Mysp3/4		Number of samples
			Number of Haplotypes	Accession numbers	Number of Haplotypes	Accession numbers	
<i>Antrozous pallidus</i>	Anpa	AY395892 AY395878	1		1		23
<i>Artibeus jamaicensis</i>	Arja	AF263226					0
<i>Choeronycteris mexicana</i>	Chme	AY395808					0
<i>Corynorhinus rafinesquii</i>	Cora	AF326091	3	KX755646-8		KX755758-60	3
<i>Corynorhinus townsendii</i>	Coto	AY395893 AY395877	1	KX755649		KX755761	16
<i>Eptesicus fuscus</i>	Epfu	AF326092 AY395894 AY395879	3	KX755650-2		KX755762-3	21
<i>Euderma maculatum</i>	Euma		1	KX755653		KX755764	15
<i>Idionycteris phyllotis</i>	Idph	AF326094	2	KX755654-5			14
<i>Lasiomycteris noctivagans</i>	Lano	AY395896 AY395881			1	KX755775	12
<i>Lasiurus blossevillei</i>	Labl	AY495479	2	KX755656-7	2	KX755765-6	2
<i>Lasiurus borealis</i>	Labo		4	KX755658-61	3	KX755767-9	4
<i>Lasiurus cinereus</i>	Laci	AY395895 AY395880	3	KX755662-4	3	KX755770-2	10
<i>Lasiurus ega</i>	Laeg	AY495483					0
<i>Lasiurus intermedius</i>	Lain		2	KX755665-6	2	KX755773-4	2
<i>Lasiurus seminolus</i>	Lase	AY495484	3	KX755667-9	3	KX755776-8	3
<i>Lasiurus xanthinus</i>	Laxa	AY495485	2	KX755670-1	2	KX755779-80	2
<i>Leptonycteris yerbabuenae</i>	Leye	AY395814					0
<i>Molossus molossus</i>	Momo	AF263215					0
<i>Mormoops megalophylla</i>	Mome	AF263220					0
<i>Myotis austroriparius</i>	Myaus	AY495493	4	KX755672-5	4	KX755781-4	7
<i>Myotis californicus/citlotebrum/leibii/melanorhinus</i>	Myca/ci/le	AY395903 AY395888 AY395904 AY395889	6	KX755676-7, KX755692-3	6	KX755785-8, KX755799-800	39
<i>Myotis evotis/thysanodes/keenii/auriculus/lucifugus/carissima</i>	Myev/th/ke/lu/ca	AY426189 AY426190 AY395901 AY395886 AY395890 AY395876	10	KX755678-86, KX755691, KX755697, KX755706	13	KX755789-92, KX755797-8, KX755801-3, KX755839-42	130

Table 1 (continued)

Species	Abbreviation	Sequences from GenBank	Mysp1/2		Mysp3/4		Number of samples
			Number of Haplotypes	Accession numbers	Number of Haplotypes	Accession numbers	
<i>Myotis grisescens</i>	Mygr		4	KX755687-90	4	KX755793-6	6
<i>Myotis lucifugus alascensis</i>	Mylu.al		8	KX755694, KX755696, KX755698, KX755700, KX755702, KX755708, KX755711-2	8	KX755804-5, KX755809, KX755812-6	64
<i>Myotis lucifugus lucifugus</i>	Mylu.lu		10	KX755695, KX755699, KX755701, KX755703-5, KX755707, KX755709 – 10, KX755713	8	KX755806-8, KX755810-1, KX755817-9	74
<i>Myotis septentrionalis</i>	Myse		2	KX755714	2	KX755820-1	53
<i>Myotis sodalis</i>	Myso		18	KX755715-32	17	KX755822-38	33
<i>Myotis volans</i>	Myvo	AY395897 AY395882	5	KX755738-42	4	KX755848-51	33
<i>Myotis yumanensis/velifer</i>	Myyu	AY395898 AY395883	10	KX755733-7, KX755743-7	9	KX755843-7, KX755852-5	189
<i>Nycticeius humeralis</i>	Nyhu	AF326102	4	KX755748-51	4	KX755856-9	4
<i>Nyctinomops femorosaccus</i>	Nyfe	AY495458					2
<i>Parastrellus hesperus</i>	Pahe	AY395899 AY395884					12
<i>Perimyotis subflavus</i>	Pesu	AY495523	6	KX755752-7	5	KX755860-4	6
<i>Tadarida brasiliensis</i>	Tabr	AF203774					13

“Sequences from GenBank” gives accession numbers for species we did not already have in our database or were poorly represented from which we designed probes. For each portion of the 16S gene we used (“Mysp1/2” and “Mysp3/4”), we provide the number of haplotypes and GenBank accession number for the sequences in our database from which we designed the probes. “Number of samples” indicates how many samples we had at our disposal to test the probes we designed. Species with a zero in this column are those for which we had no DNA to test on the array

biotinylated primers, allowing for the isolation of the 5'Cy3 modified strand. Finally, single-stranded DNA was cleaned using the Qiaquick PCR Clean-up protocol.

Pre-hybridization

Microarray slides were primed for 20 min at 55 °C in 50 mL 2 M SSC/0.2 % SDS. Slides were washed for 5 min in room temperature 0.2 M SSC, and then for 3 min in room temperature ddH₂O. Slides were centrifuged at 2000 rpm for 3 min and incubated at 50 °C for 10 min. A pre-hybridization solution [10 µl of 55 °C Vial 7 (Genosphere), 1 µl Human Cot-1 DNA[®] (1 mg/ml, Invitrogen; heated to 96 °C, then on ice), and 9 µl nuclease free water for each slide to be hybridized] was incubated at 80 °C for 10 min and was then maintained at 50 °C. Slides were pre-hybridized at 50 °C for 2 h with 18 µl of the pre-hybridization solution. Slides were then washed three times (15 min in 65 °C 2 M SSC, 0.2 % SDS; 10 min in room temperature 2 M SSC; 10 min in room temperature 0.2 M SSC). Slides were centrifuged at 2000 rpm for 3 min and then remained in a 42 °C oven until needed for hybridization.

Hybridization

Eleven µl of hybridization buffer [10 µl of 55 °C Vial 7 (Genosphere), 1 µl of Human Cot-1 DNA[®] (1 mg/ml, Invitrogen; heated to 96 °C, then on ice) for each microarray to be hybridized] was added to a microcentrifuge tube with 8 µl of the prepared sample and 1 µl of the positive control (primers Mysp 1 and 3 with a 5'Cy3 modification diluted 1:1000). The sample was then incubated at 80 °C for 10 min, and was then held at 42 °C. The sample was hybridized at high stringency 42 °C for at least 18 h. Post-hybridization washes (15 min in 65 °C 2 M SSC, 0.2 % SDS, 10 min in room temperature 2 M SSC, 10 min in room temperature 0.2 M SSC) were done under foil tents to protect the slides from light exposure. A GenePix 4000B, running GenePix-Pro 4.1 software, was used to scan and visualize the post-hybridization slides. The photomultiplier tube (PMT) was set at 600 for the 532 nm (Cy3) channel.

Microarray validation

Single species

To determine the species specificity of each probe, we hybridized DNA from each of the 30 species/species complexes for which we had DNA samples available to individual microarray slides containing all 201 probes, resulting in a total of 30 test slides (one for each species/complex). For each species, a combination of PCR product from one individual bat per haplotype per species was hybridized to

the microarray. Any probe fluorescing at 30 times or greater above background fluorescence was considered a positive result. After all species data were acquired, any probes with false negative results (probes that failed to hybridize with DNA from the target species) were removed (Table S1). In some cases where a given species had multiple probes in a true positive profile a single probe exhibiting infrequent false positives was retained (see Table S2).

Limit testing

We tested the efficacy of the microarray to detect the presence (1) of multiple species at the same time and (2) of a single rare species in a mixed-species sample. To address the first question, we combined equal volumes of DNA for two combinations of three species: (a) *Antrozous pallidus*:*Tadarida brasiliensis*:*Parastrellus hesperus* and (b) *Corynorhinus townsendii*:*Myotis sodalis*:*M. septentrionalis*. Two PCR reactions were performed for each species combination for each labeled primer set (Mysp1/2 and Mysp3/4). These PCR products were combined during the strand separation step; the DNA was processed and two slides (one for each species combination) were hybridized as described above. Second, we tested the ability of our array to detect rare species in a mixed-species sample. To do this, we tested the array with DNA from PCR reactions containing a 1:500 ratio of DNA from two species (*A. pallidus*:*C. townsendii*, and *Lasiurus blossevillei*:*Myotis grisescens*). Additionally, we tested 1:500 ratios (0.001 g:0.5 g) of guano from known individuals mixed together; these species combinations were *Eptesicus fuscus*:*A. pallidus* and *E. fuscus*:*Lasiurus cinereus*. All guano came from individual bats captured in the field or from captive animals (in the case of *A. pallidus* and *L. cinereus*) so species identity of the guano pellets was certain. The entire 0.5001 g sample for each species combination was used for DNA extraction, amplification, and hybridization using the protocol outlined above.

Mixed-species field-collected guano testing

An individual guano pellet weighs ~0.01 g. Approximately 2–10 g of guano (roughly 200–1000 fecal pellets) collected from putatively mixed-species roosts in two geographically separate localities were also tested against the microarray. We extracted DNA from two to three ~0.3 g subsamples as described above. In addition, ten individual pellets from each roost were sequenced to validate the results of the microarray analysis using previously outlined methods (Zinck et al. 2004). The roosts in Location 1 were in Indiana and were a part of a larger study monitoring the endangered Indiana bat, *Myotis sodalis* (M. Holman, unpublished data) and were suspected to house *M. sodalis* and *M. lucifugus*. Artificial roost boxes were mounted on poles and placed in

a field or forest edge. Guano was allowed to accumulate on tarps placed under the roost boxes and collected at irregular intervals. The roosts in Location 2 were in Arizona and were monitored as a part of a larger study to determine roosting preferences of southwestern bats (Wilhelm 2012). Tarps were spread under bat boxes installed under bridges to collect guano over a 24-h period. Screens installed under “bat barks” placed on Ponderosa pine trees collected guano and were checked every 30 days (Wilhelm 2012). None of the investigators had prior knowledge of the species utilizing the roosts.

Results

Validation of species-specific probes and controls

We designed and tested 201 probes and 2 positive controls (Table S1). The individual species tests resulted in 74 true positives (hybridized to target species only), 83 false negatives (did not hybridize to target species), and 24 false positives (hybridized to target species and non-target; Tables S1 and S2). Twenty probes were designed for species for which we had no DNA for testing, but these probes did not fluoresce for non-target species (Tables S1 and S2). There was no discernible differentiation between the GC content and melting temperature (T_m) of probes that successfully hybridized to the target species and those which failed to hybridize (Fig. 1). Probes that were rejected due to excessive cross-hybridization had melting temperatures above 54°C and GC contents above 33%. Our current array, with

109 probes, contains between 1 and 8 probes per species or species complex (Table S2).

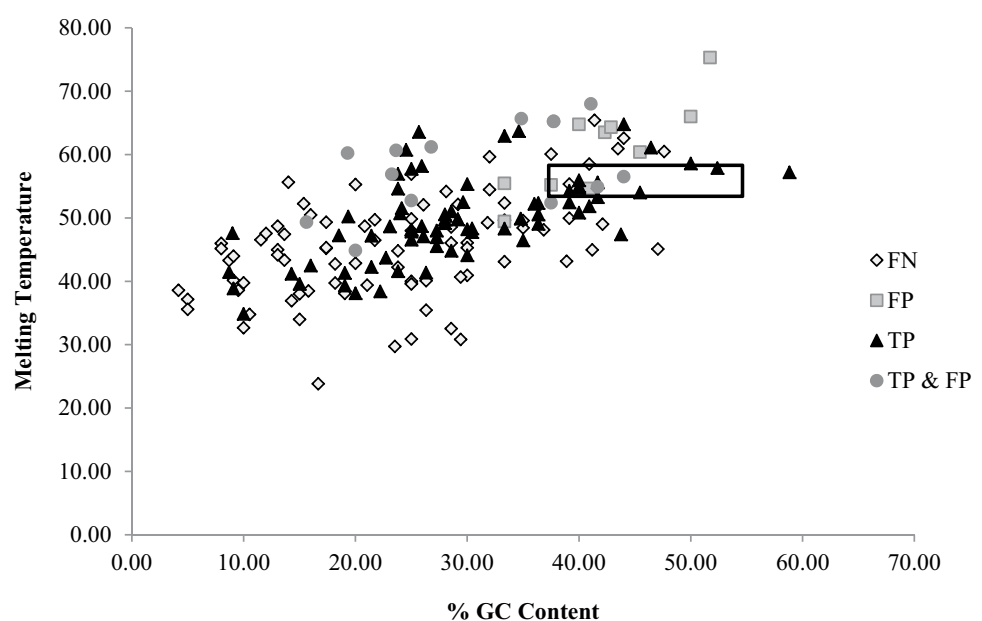
Limit testing

We detected all species present in the samples containing mixed DNA from three species; all species specific probes fluoresced above background. When using DNA and our mixed-species guano extraction protocol, our array has a limit of detection of at least 1:500. In our test dilutions both species' probes fluoresced at >30 times above background.

Mixed-species field-collected guano testing

In both tests using field-collected guano our array was able to detect multiple species present within a guano sample. When the species identifications from individual pellets were compared to the microarray species composition, the two methods detected both species for roosts in Location 1 (*M. lucifugus lucifugus*, and *M. sodalis*). For roosts in Location 2, more species were identified when using the microarray than from sequencing individual pellets. Sequences for Location 2 were of poor quality, possibly due to DNA degradation as a result of direct exposure of the samples to sunlight at the collection site, and only *A. pallidus* was identified through individual guano pellet sequencing (Wilhelm 2012). The microarray analysis for this site, on the other hand, had clear signals well above the 30 times threshold, and identified *A. pallidus*, *E. fuscus*, *L. cinereus*, and *T. brasiliensis*.

Fig. 1 Percent GC content and melting temperature (T_m) of the probes designed in this study, showing the false positive, false negative, and true positive probes. The box indicates the ideal GC content and T_m for probe function (Pfundner et al. 2004) but the spread of the data points on the graph indicates that these factors were not important to the specificity of these probes. FN false negative, FP false positive, TP true positive, TP and FP a probe that fluoresced for the target species and one or more additional species



Discussion

We designed species-specific DNA probes and a microarray capable of identifying 39 of the 47 named bat species in the US and Canada, including all 14 species on the microarray designed by Lung et al. (2013), using degraded, mixed-species DNA sources. The resulting microarray is very sensitive, capable of detecting even tiny amounts of DNA from the species we used to test the probes. While we have only tested the utility of these probes for the microarray, we suggest that they might also prove useful for identifying species using quantitative PCR. We hope that these molecular tools will prove useful in non-invasive biodiversity studies, particularly in light of the increasing concern about the continued spread of white-nose syndrome (WNS). When stored in a dark, dry place, guano can yield amplifiable DNA at least 10 years after collection (Zielinski et al. 2007). This means our microarray could be used to identify species presence or diversity in roosts from guano collected pre-WNS infection to that collected post-WNS infection to better document changes in site occupancy and bat diversity over time in affected areas. However, there have been no studies of DNA persistence in guano under various natural conditions; such studies should be carried out to determine how quickly DNA degrades when conditions are not ideal.

While a microarray to identify Canadian bats has been recently described (Lung et al. 2013), we feel that our microarray provides a complementary tool that may prove more useful for non-invasive biodiversity monitoring. We designed our probes based on two small, highly variable segments of 16S mtDNA while Lung et al. (2013) used a much larger portion of the COI gene. Small fragments are much more likely to be amplified from degraded samples, such as guano collected from roosts, and the concentration of variation in short segments of rRNA genes rather than being spread across third codon positions in protein-coding genes such as COI allows for more efficient probe design and higher success rates (see also Kochzius et al. 2010). Further, we have designed and tested probes for a larger range of North American bat taxa, increasing the utility of our microarray for species identification and biodiversity monitoring across a wider geographic area. Our array includes probes to identify almost all federally-listed threatened or endangered bat species in the continental United States and Canada (with the exception of *Leptonycteris nivalis*), including *Corynorhinus townsendii* (endangered subspecies *ingens* and *virginianus*), *L. yerbabuenae*, *Myotis grisescens*, *M. septentrionalis*, and *M. sodalis*, as well as numerous species listed as endangered, threatened, or of special concern in individual states or provinces. Thus, our array provides biologists with a valuable tool with which to non-invasively monitor species occupancy and diversity in roosts to inform management and conservation activities.

Our microarray is a valuable tool for describing species presence with the caveat that data on specific individuals is sacrificed in the process and sequence data is not produced that could be used in downstream phylogeographic, phylogenetic, or population genetic analyses. The cost and labor savings of a microarray approach, both in the field and in the laboratory, lie in the rapid census of species presence from mixed-species samples, such as the collection of fecal matter at communal roosts when the actual organisms are absent or the pooling of fecal or tissue samples collected through other means. Direct sequencing of individual pellets or samples has significantly greater cost (Table 2) and may require considerable additional time to process samples and curate sequences. Although Rubenstein (2003) proposed that in order to be cost effective a microarray should be broadly applicable, Summerbell et al. (2005) suggested that microarray technology was not practical for groups of >200 species because all probes must be proof tested for sensitivity, specificity, and cross-reactions. Because of the relatively low numbers of potential species at a given location, microarray technology is ideal for use with vertebrate taxa in many regions. This is especially true in those cases where multiple species communally roost at or otherwise utilize the same site, such as bats and birds, and for other mixed-species non-invasive samples such as those from owl pellets, latrine sites, rodent middens, and hair traps.

The most expensive step of microarray development is the testing of probes for cross-hybridization with non-target species (Summerbell et al. 2005). Our results show the necessity of laboratory testing under applied conditions using real samples as opposed to in silico testing based on model theory. In fact, the majority of our successful (and unsuccessful) probes fell well outside the suggested ranges of GC content and melting temperatures for microarray probes (Pfunder et al. 2004 and used by other authors, such as; Hajibabaei et al. 2007) indicating that these parameters

Table 2 Approximate costs (in US dollars) associated with identifying samples using traditional Sanger sequencing (recognizing that costs vary tremendously among approaches, laboratories, suppliers, etc.) versus using a microarray

	Direct sequencing	Microarray
Consumables (tubes, gloves, reagents, etc.)	\$0.25–1.00	\$5.00
Extraction	1.50–3.00	3.00
PCR	1.00–3.00	8.00
Bi-directional sequencing	5.00–12.00	n/a
Slides	n/a	250.00
Single sample cost	\$7.75–19.00	\$266.00
Cost to identify 500 pellets	\$3875–9500	5 slides = \$1330

These estimates do not include labor costs

should be adjusted when designing probes (Fig. 1). Microarray probes simply fail to always function as predicted (see also Kochzius et al. 2010; Noyer et al. 2015) so there is a certain amount of actual trial and error that must be incorporated in the successful development of a microarray (Fig. 1, Table S1). This is particularly important because, as the number of species to be detected increases, the cost of testing probes developed for each new species against the entire pool of potential targets to validate their specificity increases exponentially. At some point, the cost of additional testing may begin to outweigh the savings associated with the use of the microarray to bulk test samples. In some cases the creation of taxa- or region-specific probes (as a means to limit the pool of target species) may be a more cost effective strategy for array design.

The savings and benefits of developing and applying a field-tested species-specific microarray to detect species presence at roost sites can not only outweigh those of other molecular techniques such as direct sequencing (Table 2), but also those associated with traditional field survey methods such as visual searches or captures (Wadley et al. 2014). For example, it can take several surveys to visually detect *Corynorhinus townsendii* roosting in a cave or mine because they often move between roosts so that there is no reliable way to predict their presence to coincide with surveys (Sherwin et al. 2000). The expense of returning to a site multiple times to conduct surveys can be high and if capture is required, the additional cost of having handling skills and rabies prophylactics must be considered. As an alternative, collection of guano can be achieved in a single visit and no special skills or prophylactics are required other than following appropriate decontamination protocols. This is particularly useful to monitor dwindling bat populations hard hit by WNS without disturbing the bats themselves.

Understanding the limits of detection for a microarray is a key factor in determining its utility in survey and management plans. We tested a relatively extreme scenario of a 1:500 ratio of two bat species in a mixed sample (roughly corresponding to one fecal pellet among 500 others), and found that both species were detected in all cases. While we did not test additional dilutions, and it is possible that our method can be used to detect even lower relative concentrations of DNA; these results suggest that our microarray is a powerful tool for detecting species presence when individual species are rare (in terms of population size or frequency of use) at a site. Given that large amounts of guano may accumulate at a site, several factors help to determine the number of microarrays necessary per sampling unit. The first consideration is whether the research question or survey and management obligations require that all species, no matter how rare in the sample, be detected or whether identifying dominant species is sufficient. Guano samples can be sub-sampled numerous times with the number of

subsamples being run determined by the goals of the study. It is important to note that although several subsamples may be pulled from the homogenized mixed-species sample, the resulting PCR products from several sub-samples may be combined for hybridization on one microarray. In the strictest cases where all species present must be detected, the average rate of successful PCR amplification per pellet of guano (approximately 70 %) must also be considered (Zinck et al. 2004) in the calculation of how many subsamples to extract and microarrays to run.

Our tests suggest that the microarray is more sensitive and is more likely to detect a greater number of species than direct sequencing of fecal pellets. Although this could be partially due to a greater quantity of DNA being hybridized to the slide than was used for sequencing, we feel this difference is also because sequencing requires template DNA of sufficiently good condition from a single individual (which can be difficult to obtain from a mixed individual guano sample) to produce readable chromatograms and alignments. Conversely, the microarray only requires template DNA to be successfully amplified via PCR, and hybridization to a probe is less stringent than sequencing with respect to DNA quantity and quality as the probes are much shorter than the length of DNA typically sequenced (Bryant et al. 2004; Forêt et al. 2007). This greatly increases the efficacy of the microarray over sequencing for samples that are degraded by time, ultraviolet light, or moisture. The mixed guano samples used to test the microarray were exposed to all of these degradation factors so obtaining good sequences was difficult, resulting in the detection of fewer species than the microarray. Direct sequencing of a few individuals will likely remain necessary for geographical regions that are underrepresented by our current sequence library to ensure that that rare species or haplotypes are indeed detectable by the probes printed on the microarray. However, as the samples from Location 2 demonstrate, the microarray performs exceptionally well even for geographical areas from which we previously had few samples.

Conclusions

We have demonstrated the utility and sensitivity of a DNA microarray for use with fecal DNA for bat species monitoring. DNA microarrays such as the one we have developed here provide an efficient method for detecting species presence using non-invasive and potentially mixed-species samples. This technology is particularly useful for detecting the presence of rare, nocturnal, elusive, protected, or otherwise difficult to study organisms, and where the handling the organism itself is undesirable or not feasible, as is the case with most of the bat species included in our array while they are in their summer and winter roosts. More broadly,

this technology is a powerful tool that can be widely used in conjunction with field studies to elucidate spatial and temporal patterns of species presence and composition at sites of interest, and provide a more cost-effective and sensitive method for biodiversity monitoring than many traditional approaches.

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